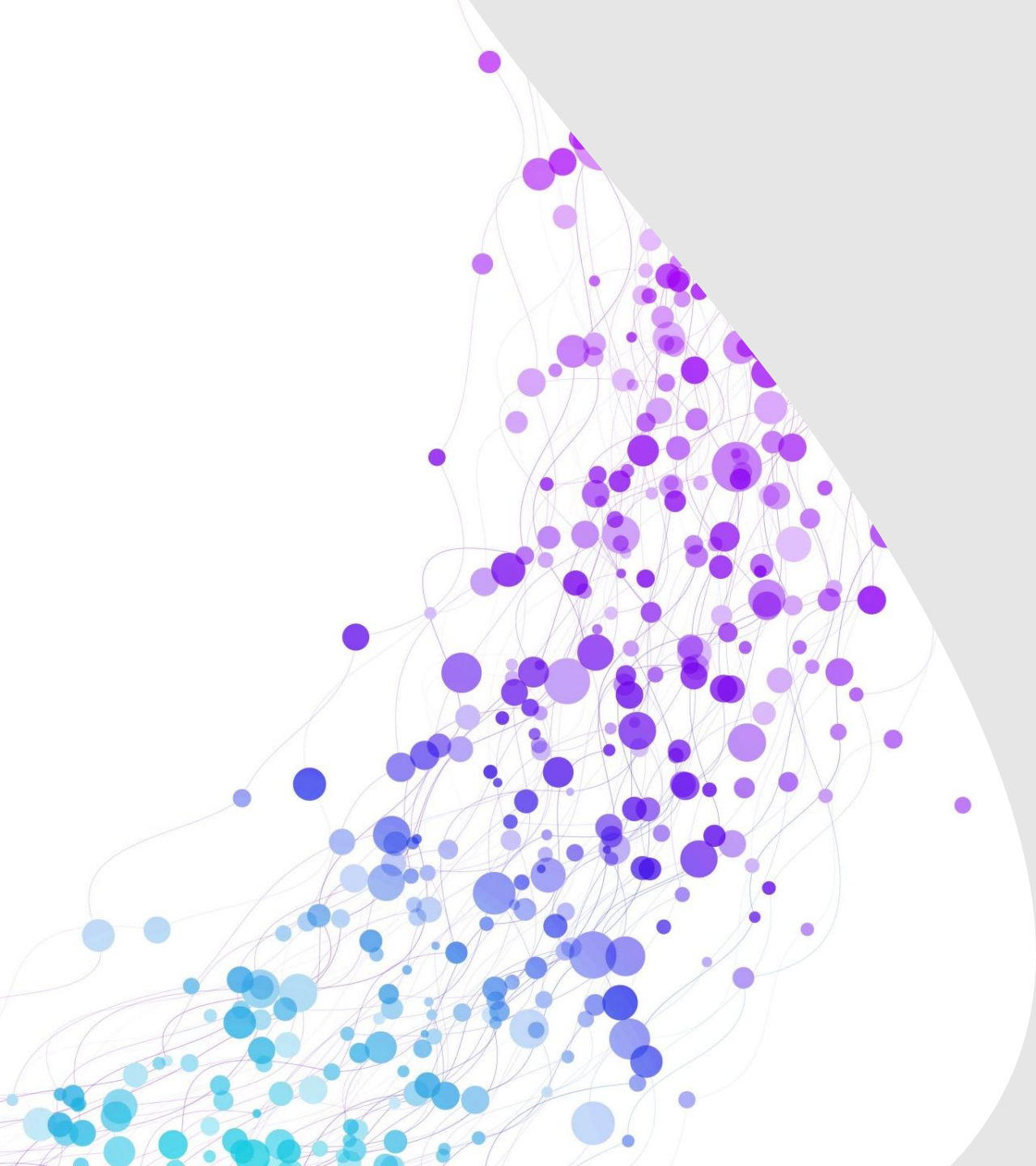
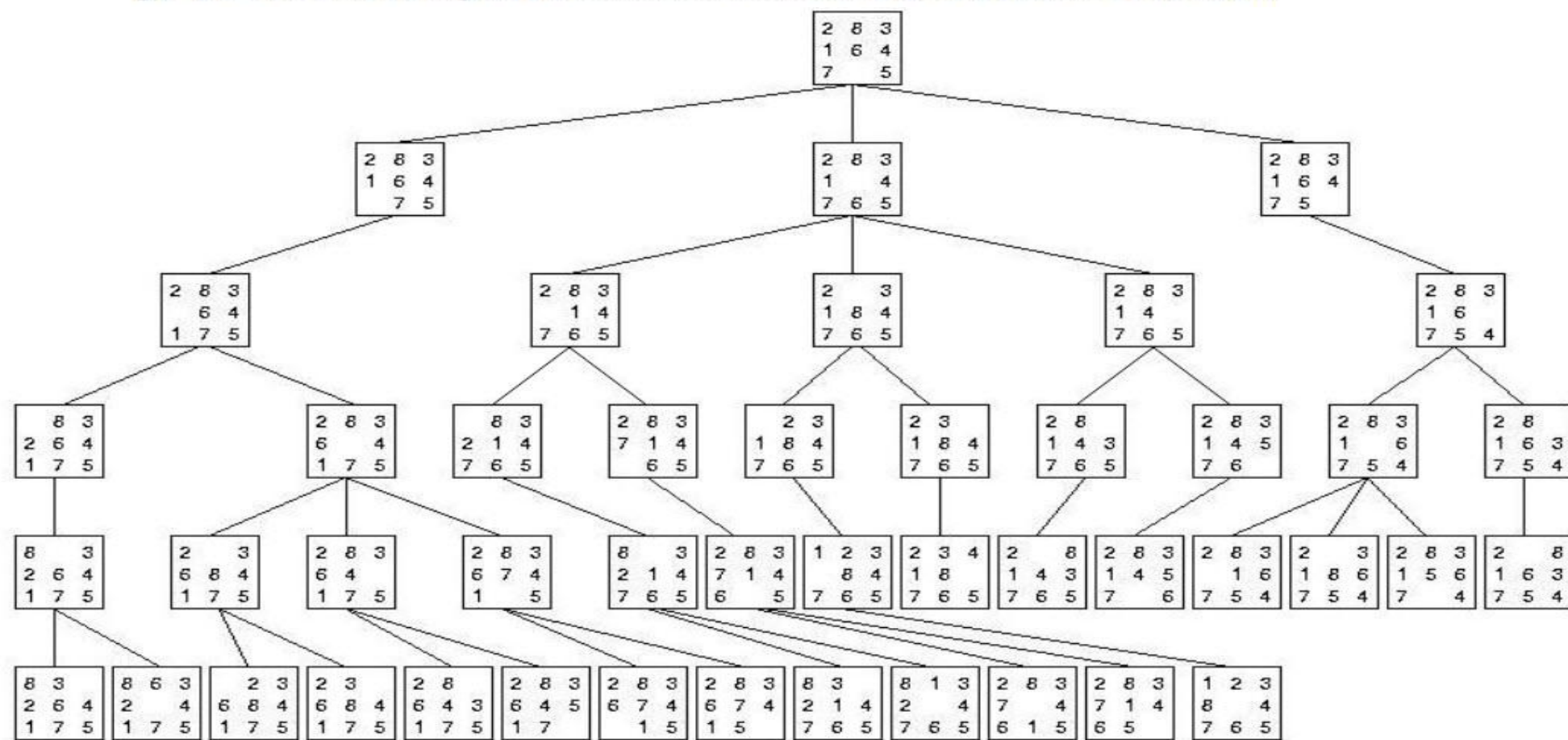




Sheet 2: Search Best First

Eng. Abdulrahman ElMadkhoun

1. The following figure gives an example showing the BFS tree for the 8 puzzle problem. List the order of the visited nodes by the following search algorithms:
 - a) Breadth first
 - b) Depth First
 - c) Limited Depth First with limit $d=3$.
 - d) Best first Search using the misplaced tiles as a heuristic function.
 - e) Best first Search using the Manhattan Distance as a heuristic function.
 - f) A* Search using the misplaced tiles as a heuristic function.
 - g) A* Search using the Manhattan Distance as a heuristic function.



BEST First Search

Misplaced Tiles

First, Calculate Heuristic using Misplaced Tiles

2	8	3
1	6	4
7		5

Initial State

1 is misplaced
2 is misplaced
3 is ok
4 is ok
5 is ok
6 is misplaced
7 is ok
8 is misplaced
So, heuristic here = $1+1+1+1 = 4$

$H(n)=4$

1	2	3
8		4
7	6	5

Goal State

$H(n) = 0$

2	8	3
1	6	4
7		5

$$1+1+1+1 = 4$$

Goal		
1	2	3
8		4
7	6	5

2	8	3
1	6	4
7		5

$1+1+1+1 = 4$

2	8	3
1	6	4
7		5

2	8	3
1		4
7	6	5

2	8	3
1	6	4
7	5	

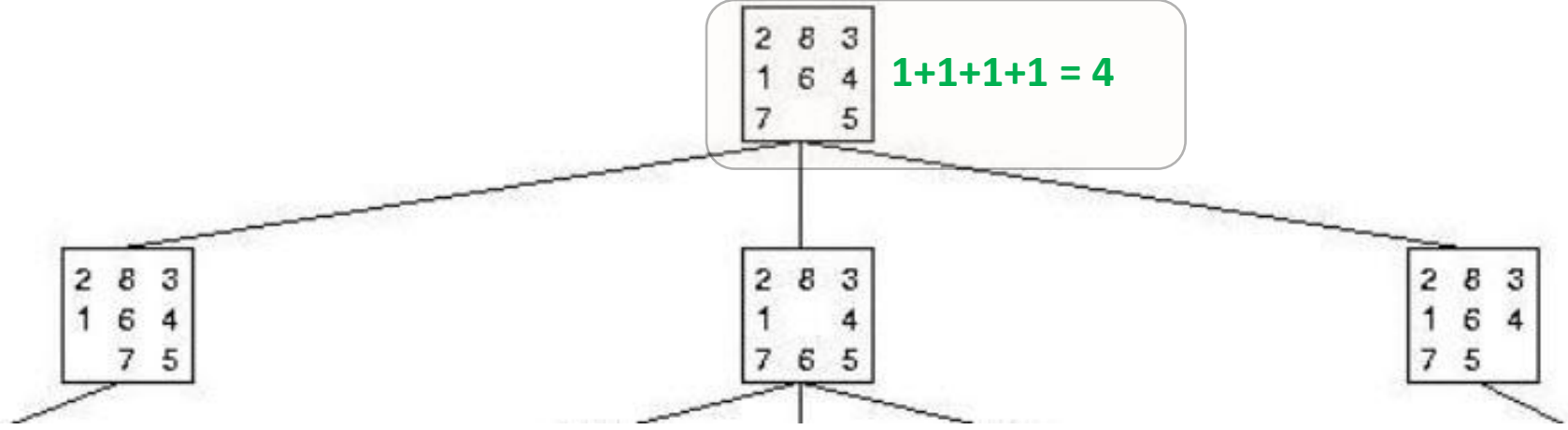
Goal

1	2	3
8		4
7	6	5

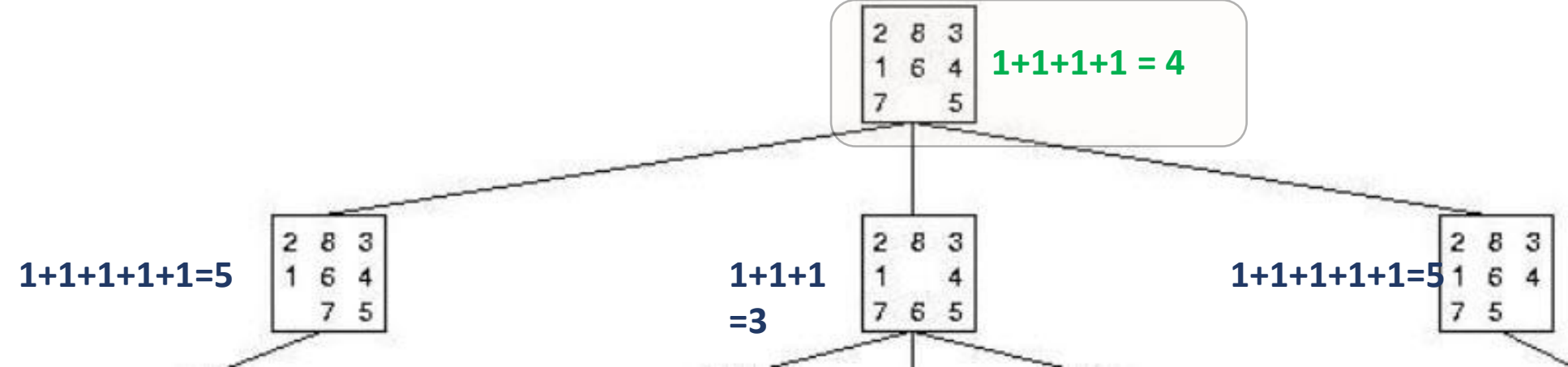
Goal

1	2	3
8		4
7	6	5

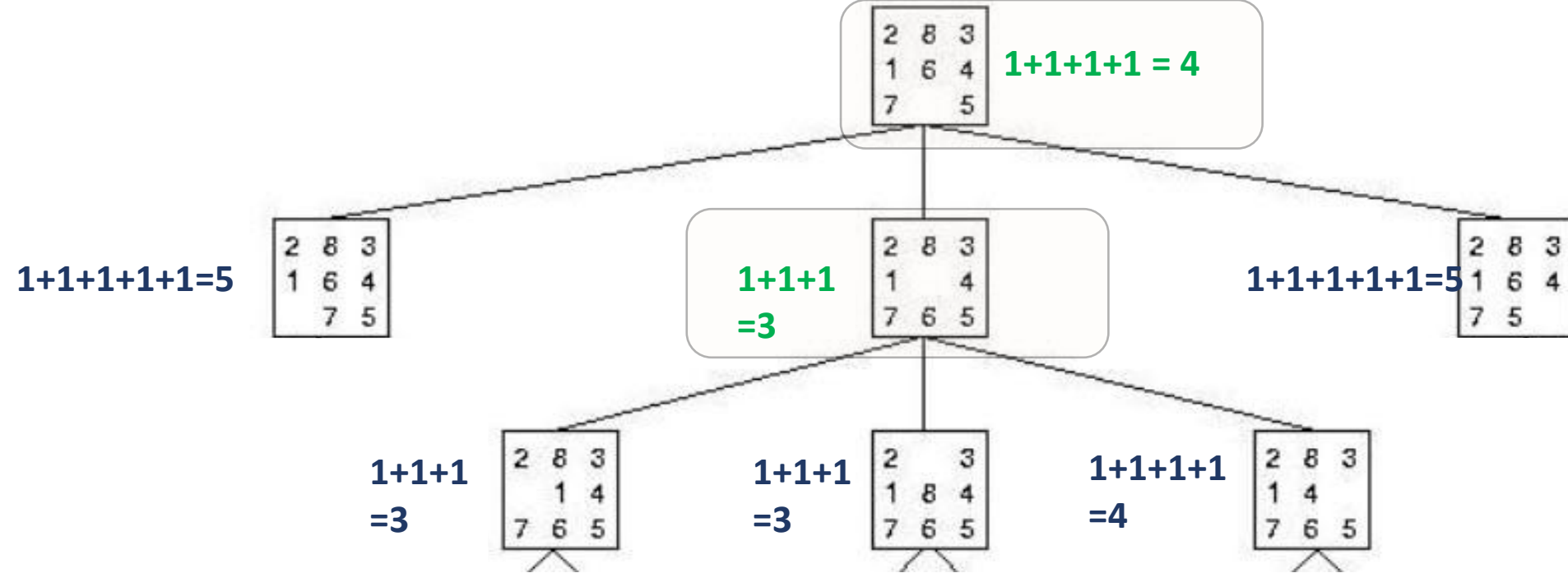
$1+1+1+1 = 4$



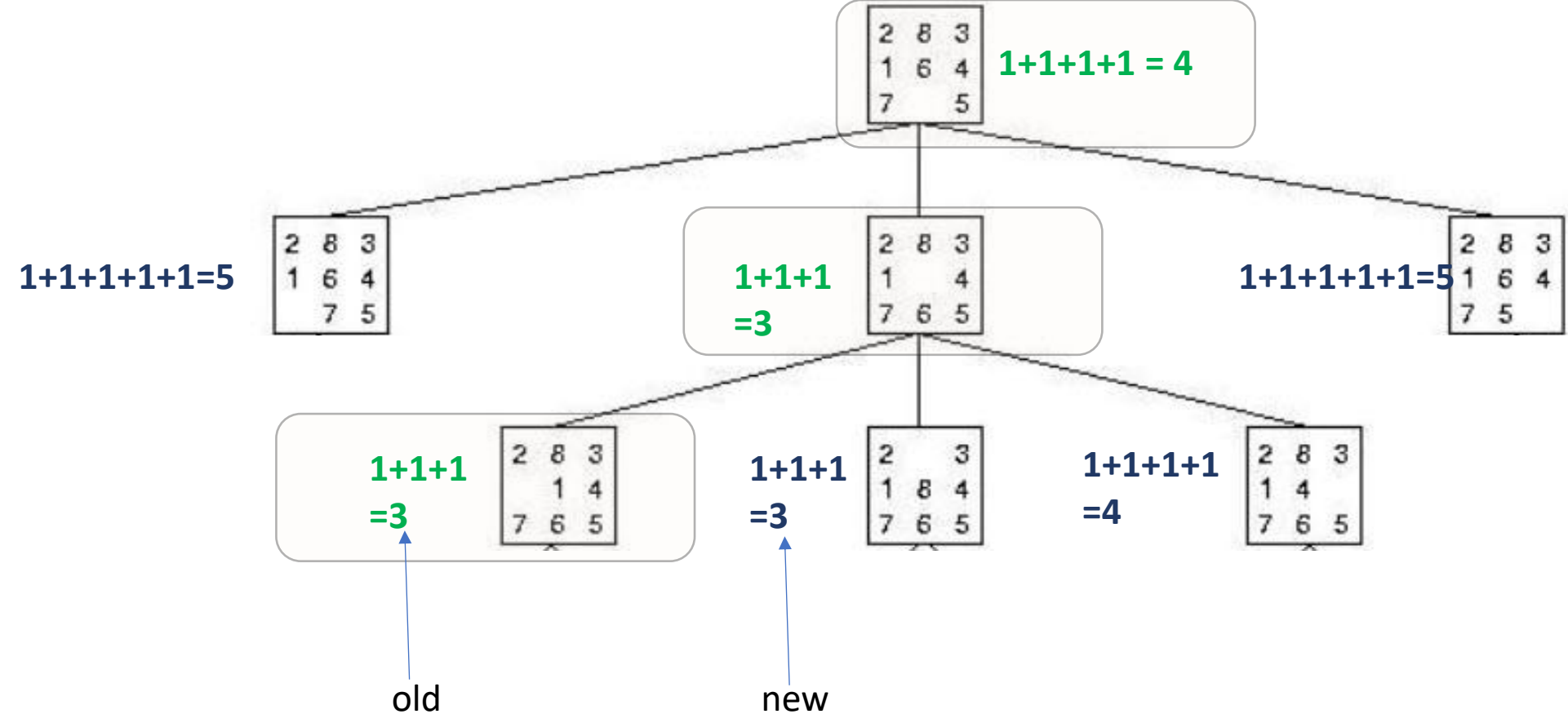
Goal		
1	2	3
8		4
7	6	5



1	2	3
8		4
7	6	5

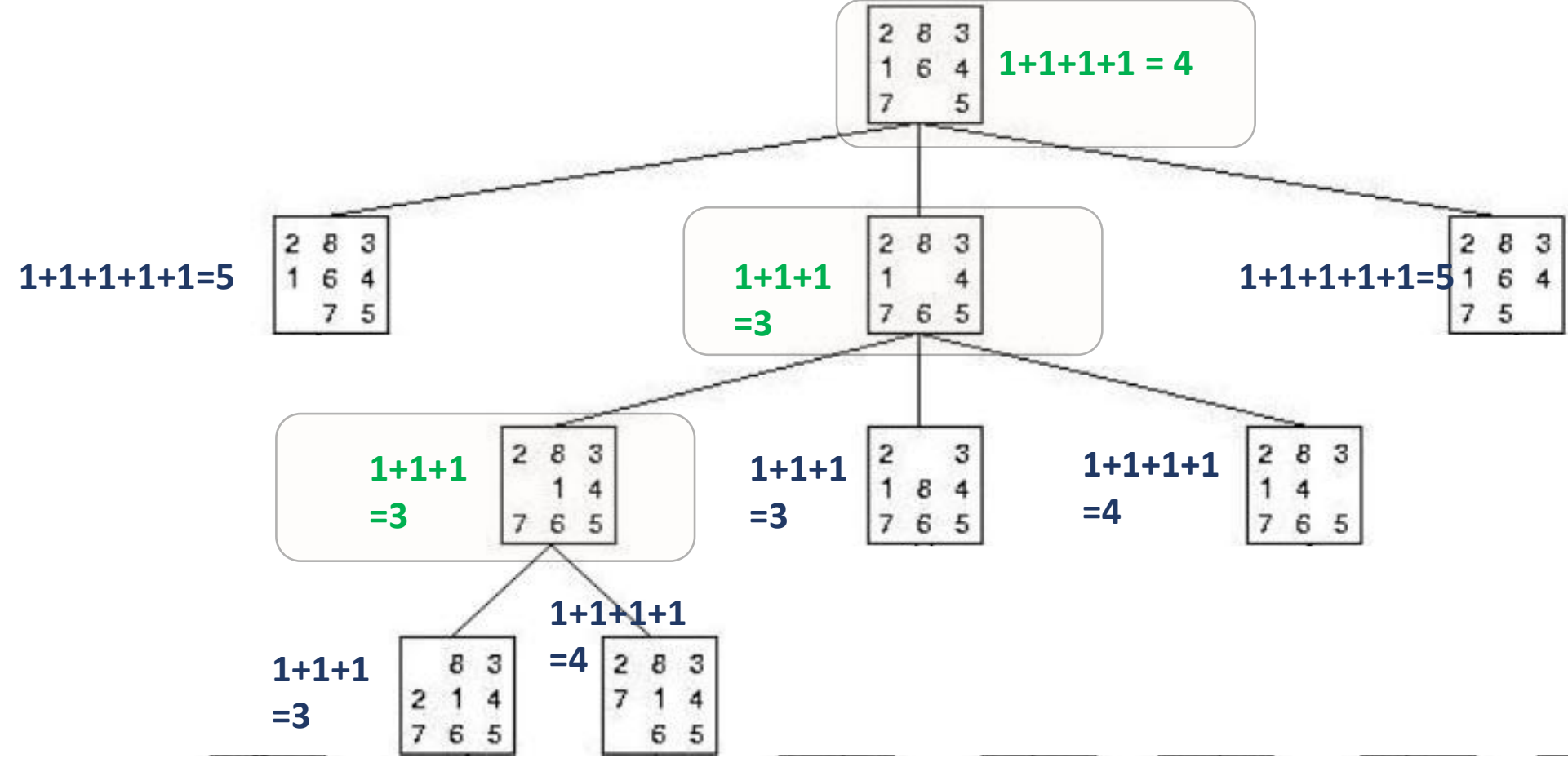


Goal		
1	2	3
8		4
7	6	5

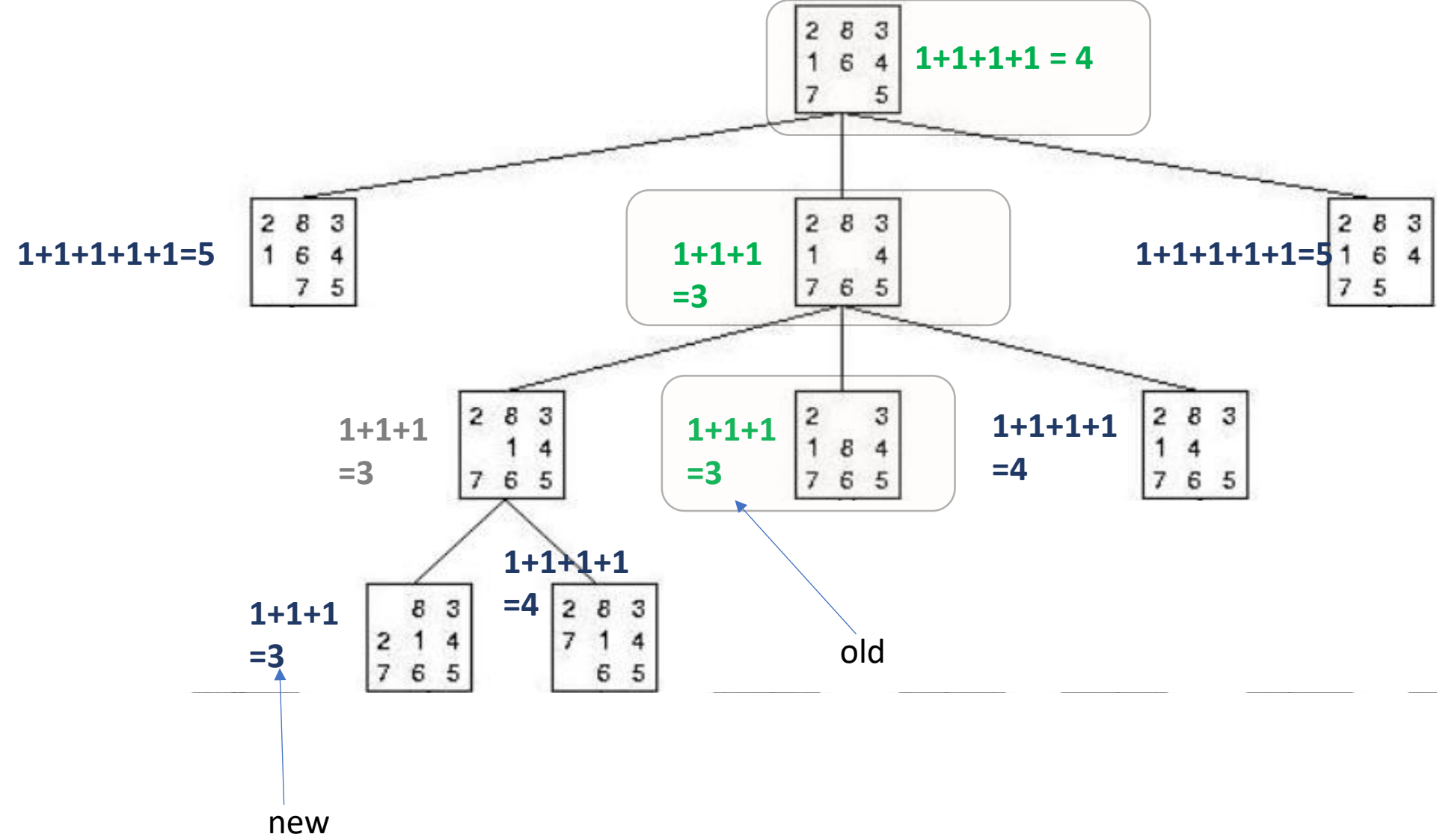


Goal

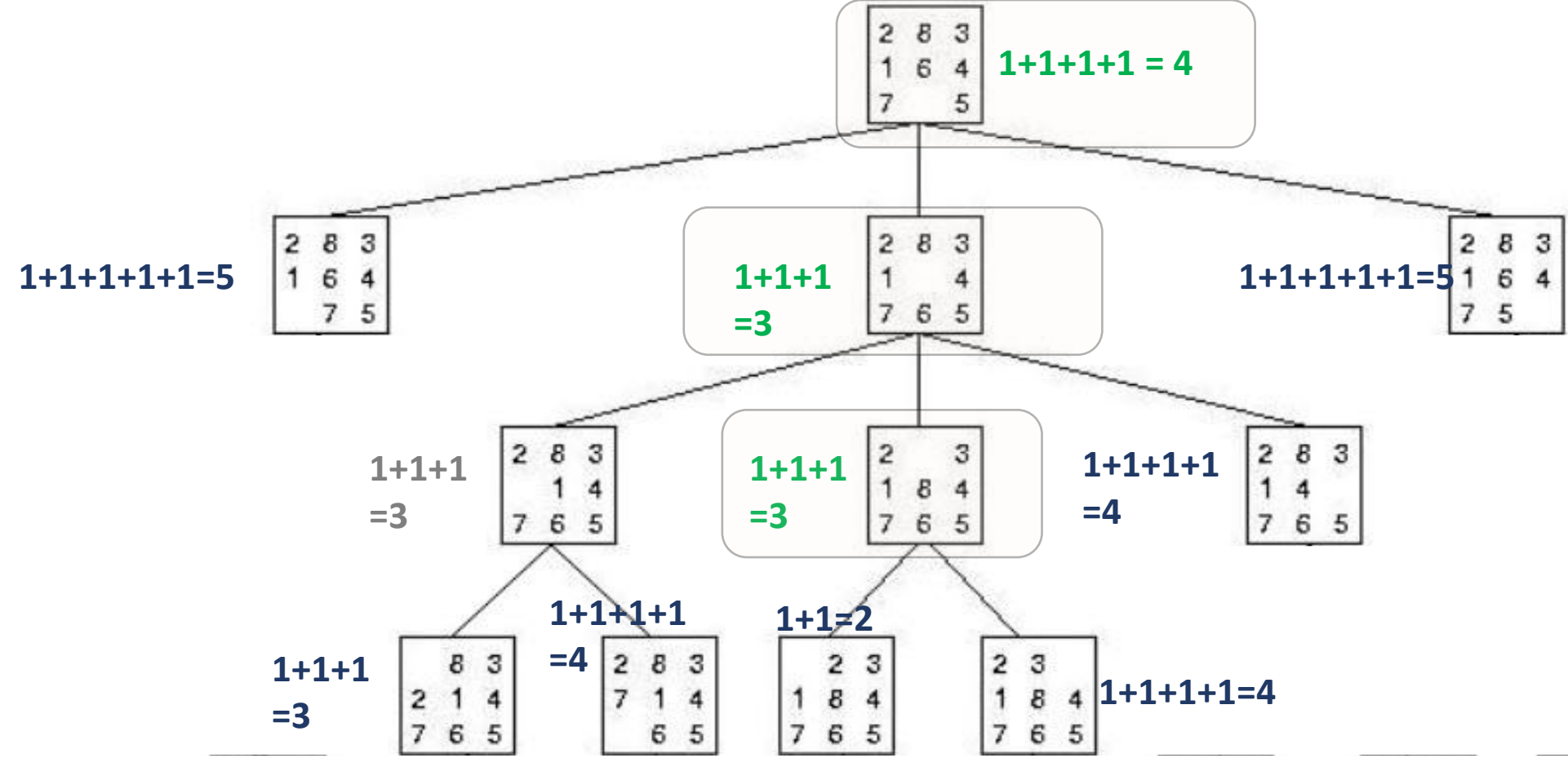
1	2	3
8		4
7	6	5



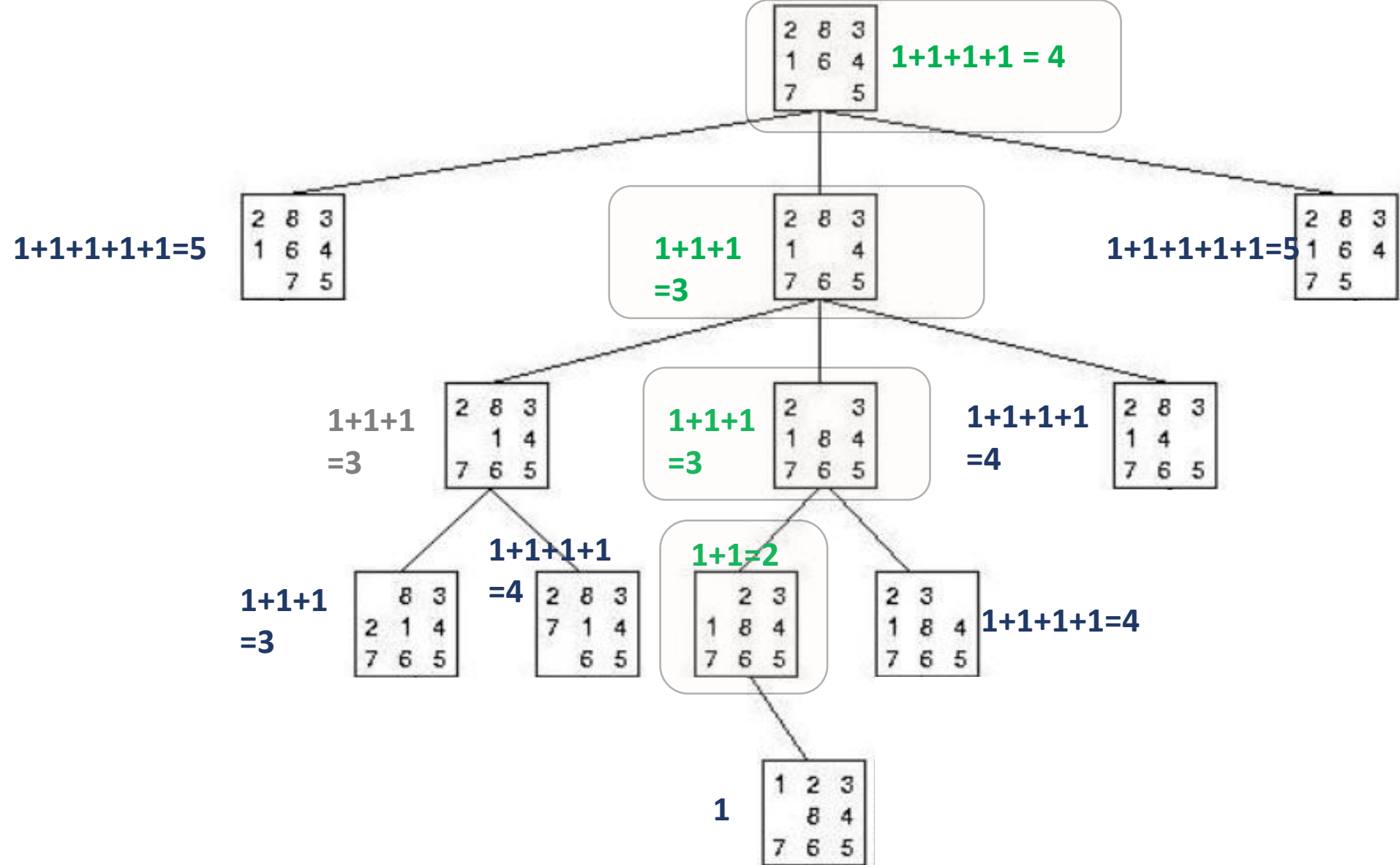
Goal		
1	2	3
8		4
7	6	5



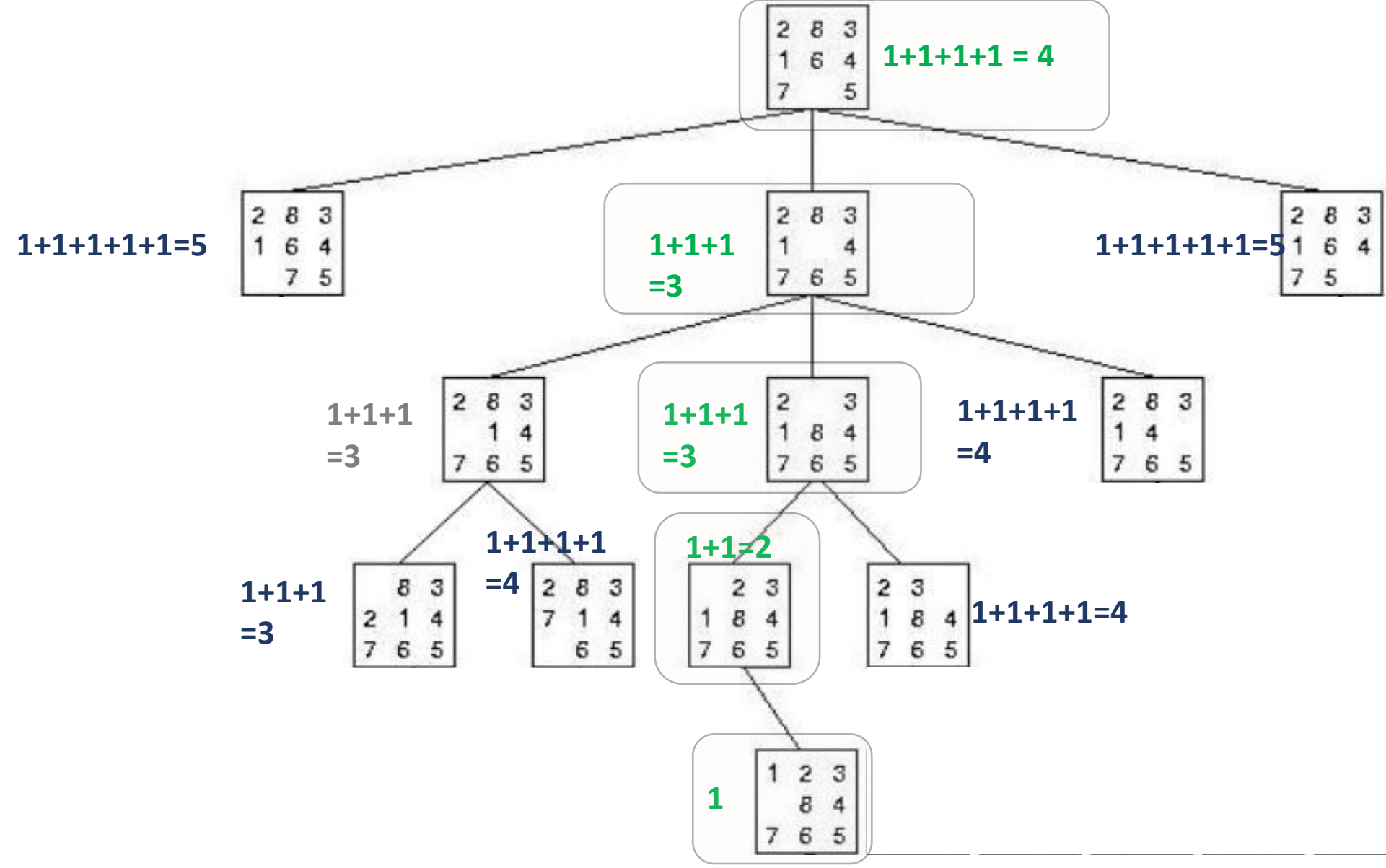
Goal		
1	2	3
8		4
7	6	5



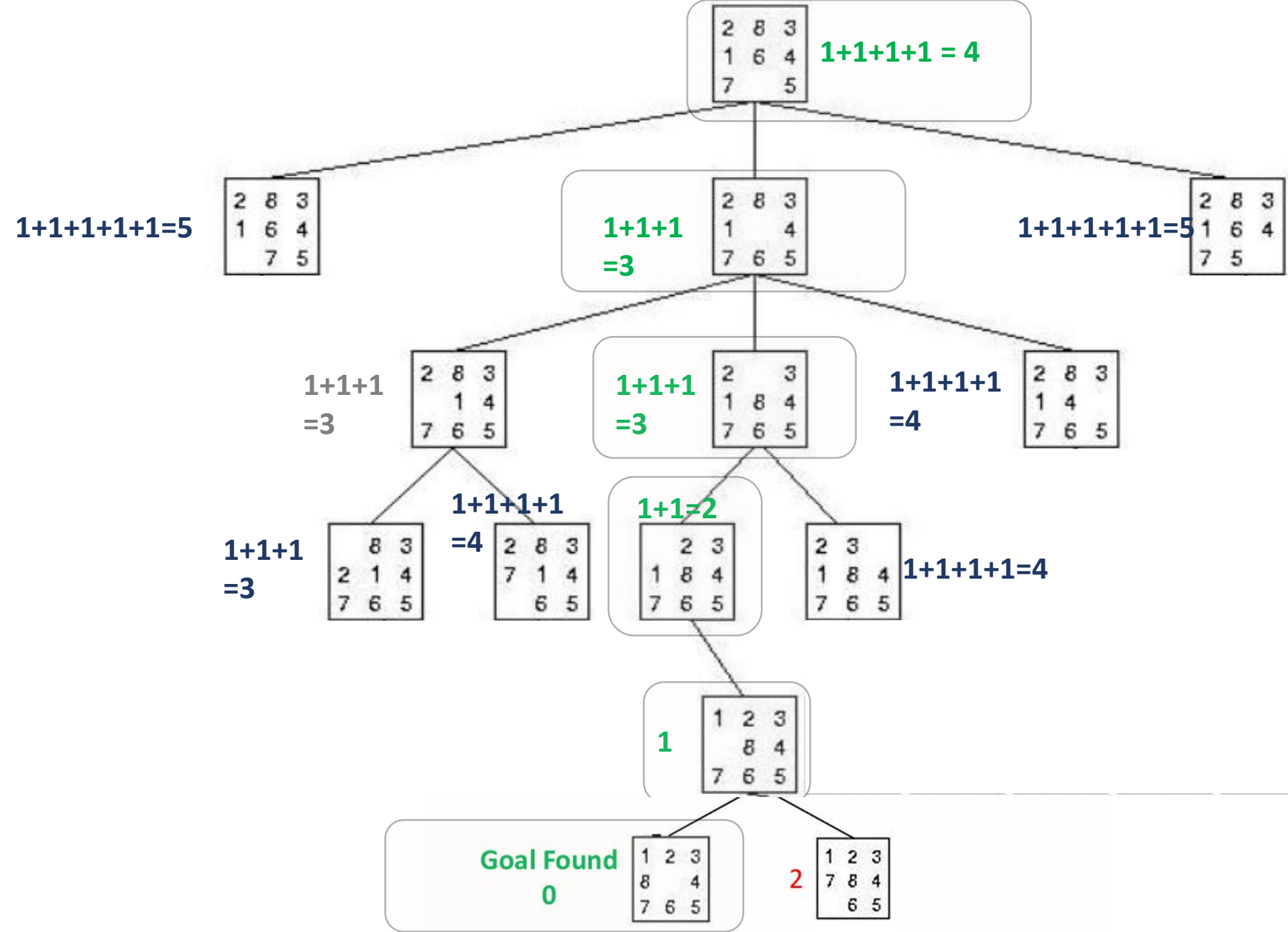
1	2	3
8		4
7	6	5



Goal		
1	2	3
8		4
7	6	5



Goal		
1	2	3
8		4
7	6	5

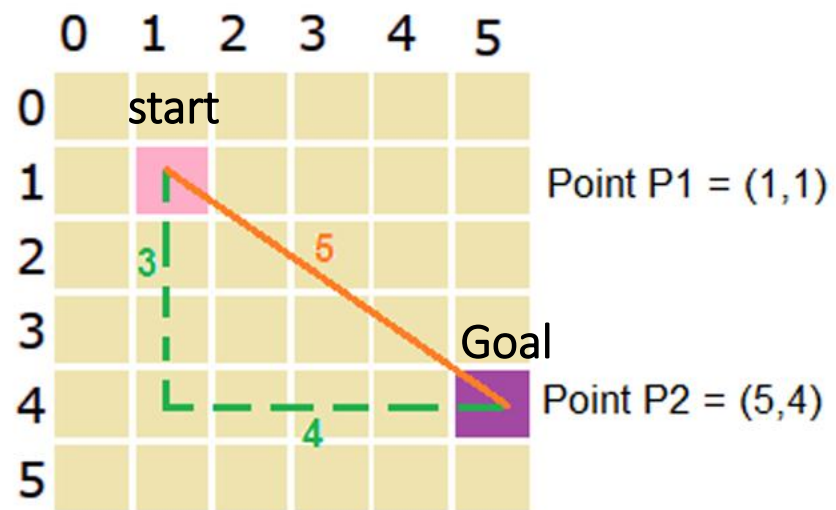


Notice that

- We choose the node with the least cost from the whole tree and if there are two nodes with the same cost (least cost) we choose the older one
- heuristic value express how similar is the current node to the goal
(The goal heuristic value = 0)

BEST First Search

Manhattan Distance



$$\text{Euclidean distance} = \sqrt{(5-1)^2 + (4-1)^2} = 5$$

$$\text{Manhattan distance} = |5-1| + |4-1| = 7$$

	1	2	3
1	2	8	3
2	1	6	4
3	7		5

We calculate the Manhattan distance for each number

For example (8)

Current coordinates (1,2) and the goal to be at (2,1)

Manhattan distance = $|2-1| + |1-2| = 2$

But for this problem for simplification, we calculate how many steps till it be in the write position

First, Calculate Heuristic using Manhattan Distance

2	8	3
1	6	4
7		5

Initial State

1 needs 1 movement to be in it's correct position
2 needs 1 movement to be in it's correct position
3 is in it's correct position
4 is in it's correct position
5 is in it's correct position
6 needs 1 movement to be in it's correct position
7 is in it's correct position
8 needs 2 movements to be in it's correct position
So, The Heuristics is = $1+1+1+2 = 5$

1	2	3
8		4
7	6	5

$$H(n) = 0$$

2	8	3
1	6	4
7		5

$$1+1+1+1+2=5$$

Goal		
1	2	3
8		4
7	6	5

Goal

1	2	3
8		4
7	6	5

2	8	3
1	6	4
7		5

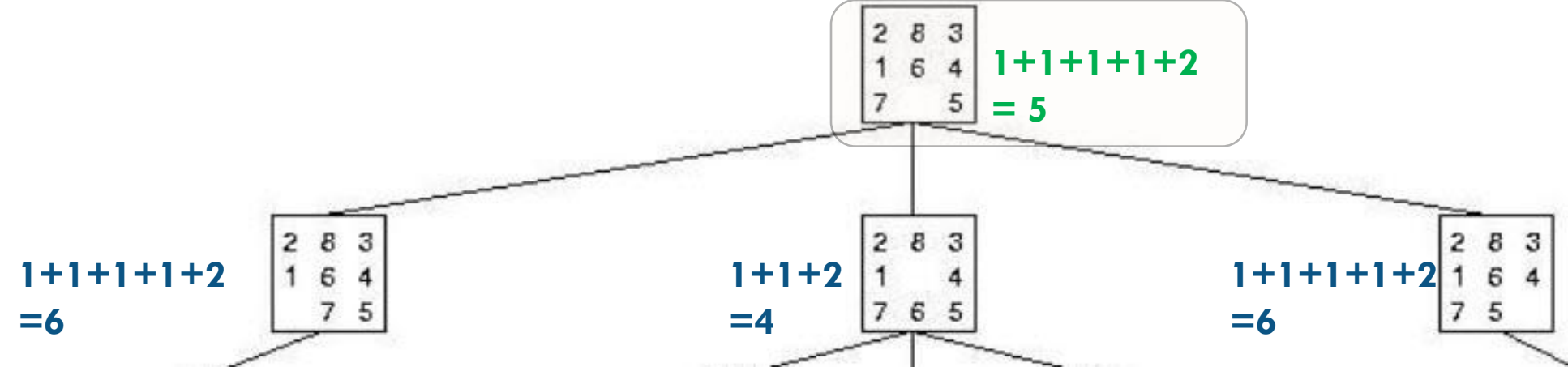
$1+1+1+1+2$
 $= 5$

2	8	3
1	6	4
7		5

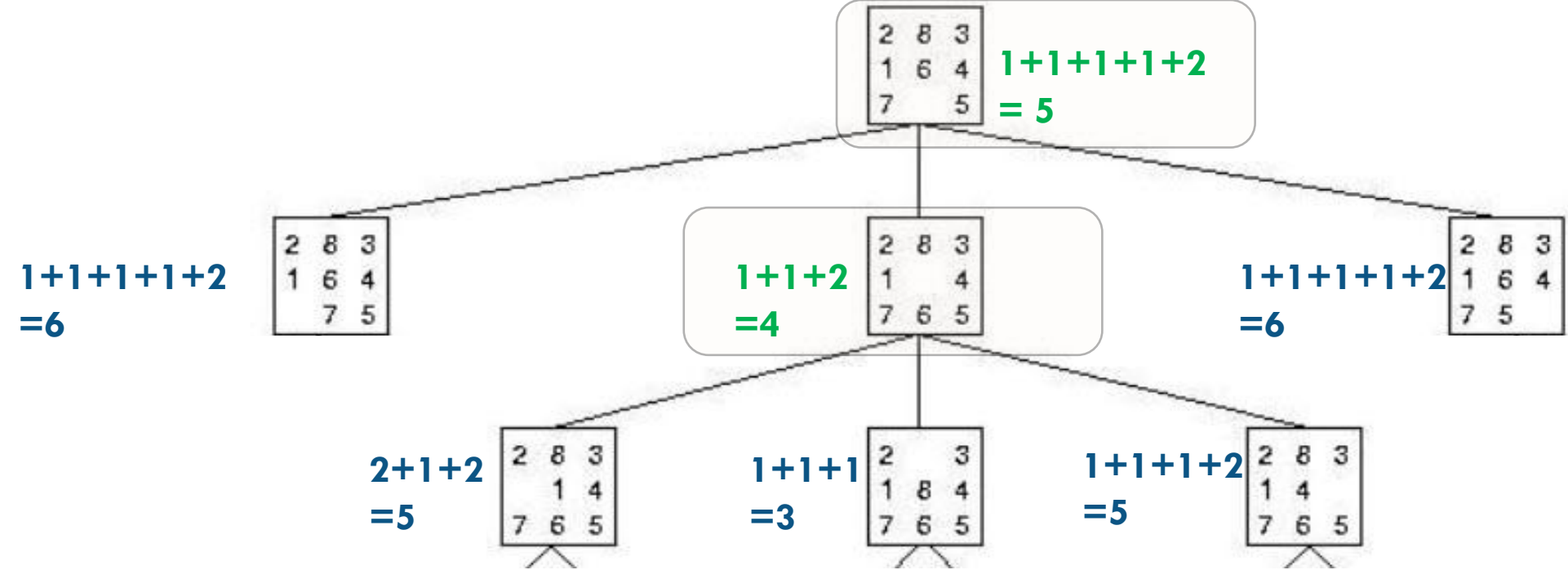
2	8	3
1		4
7	6	5

2	8	3
1	6	4
7	5	

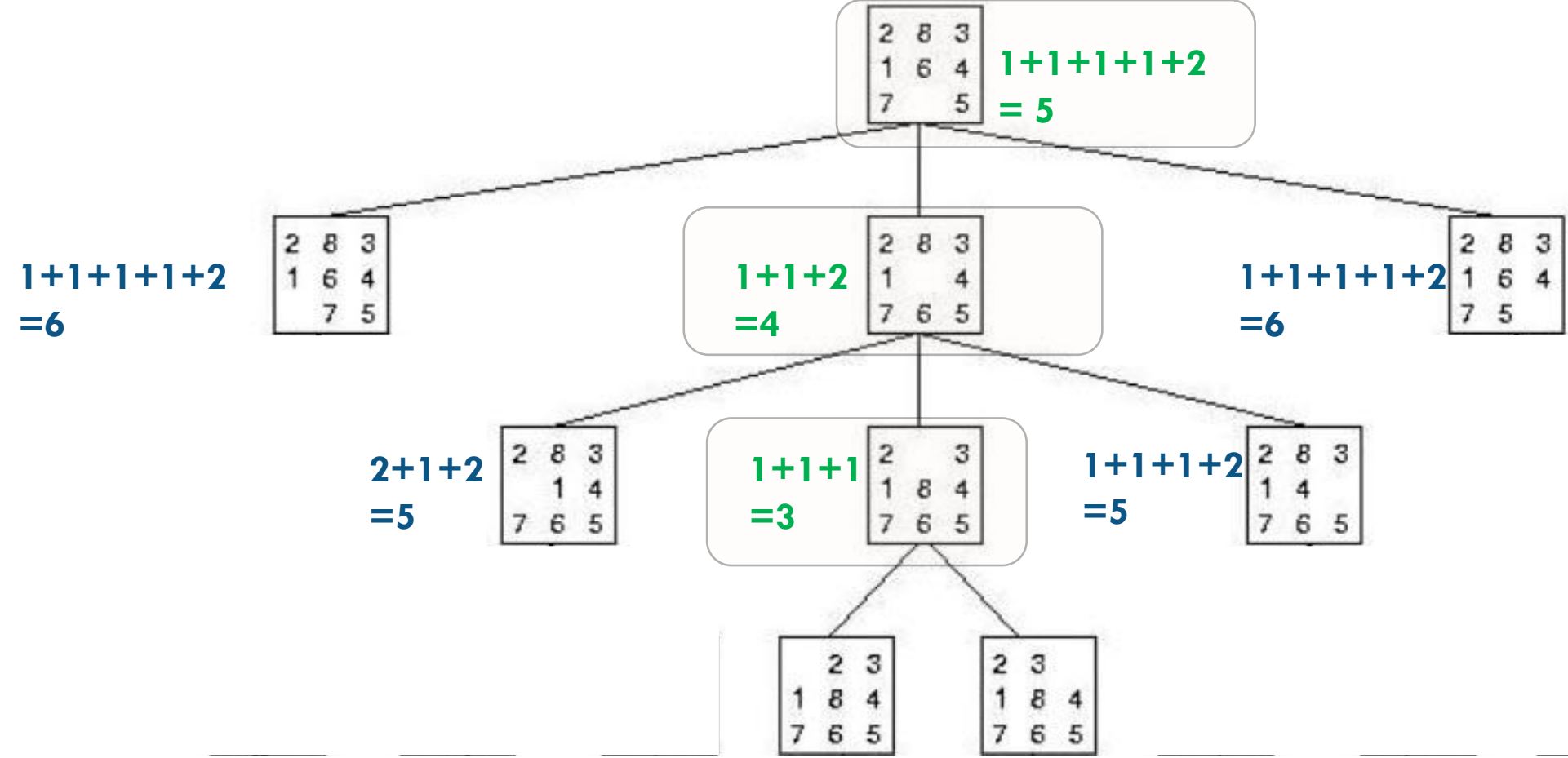
Goal		
1	2	3
8		4
7	6	5



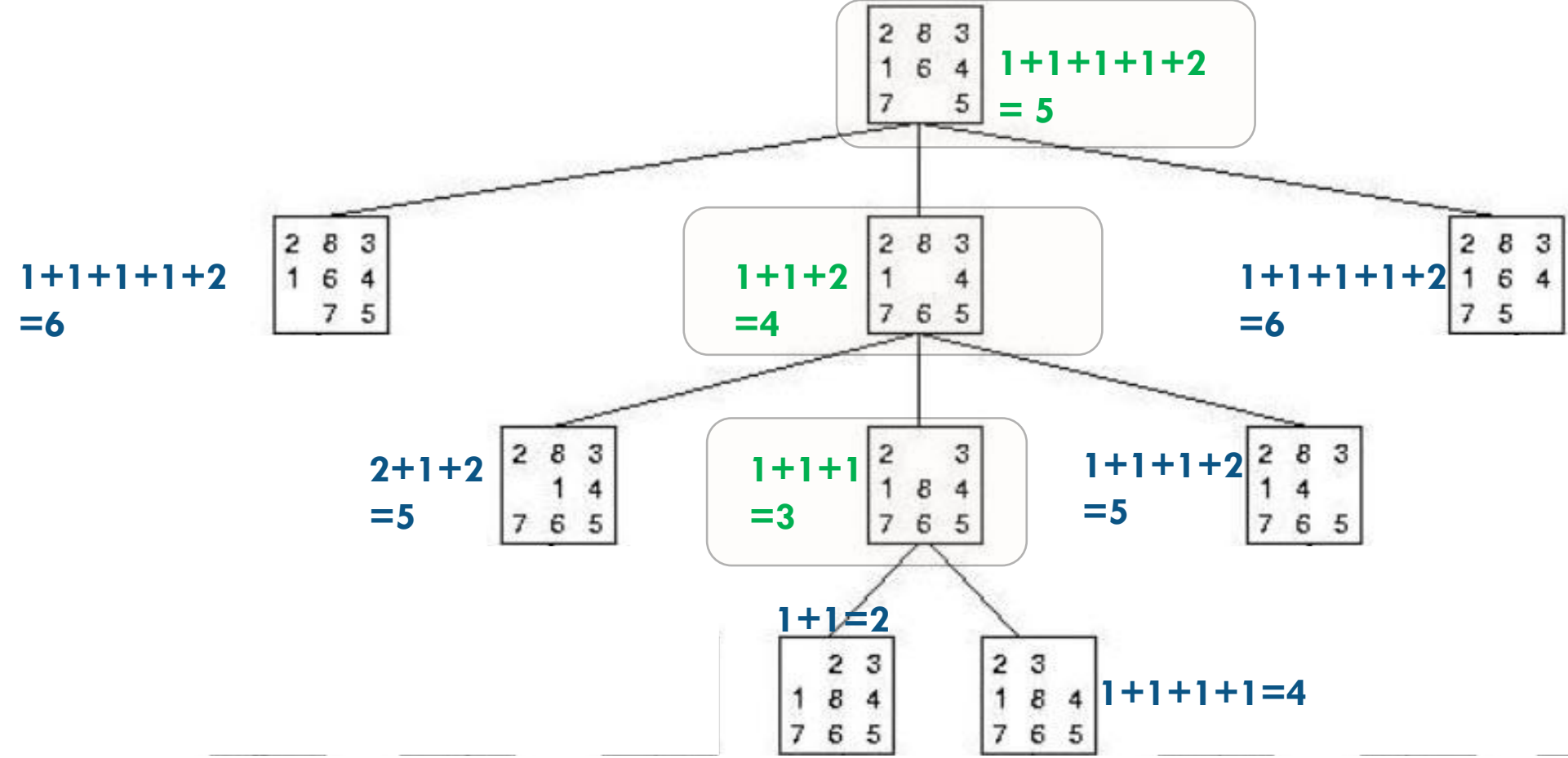
Goal		
1	2	3
8		4
7	6	5



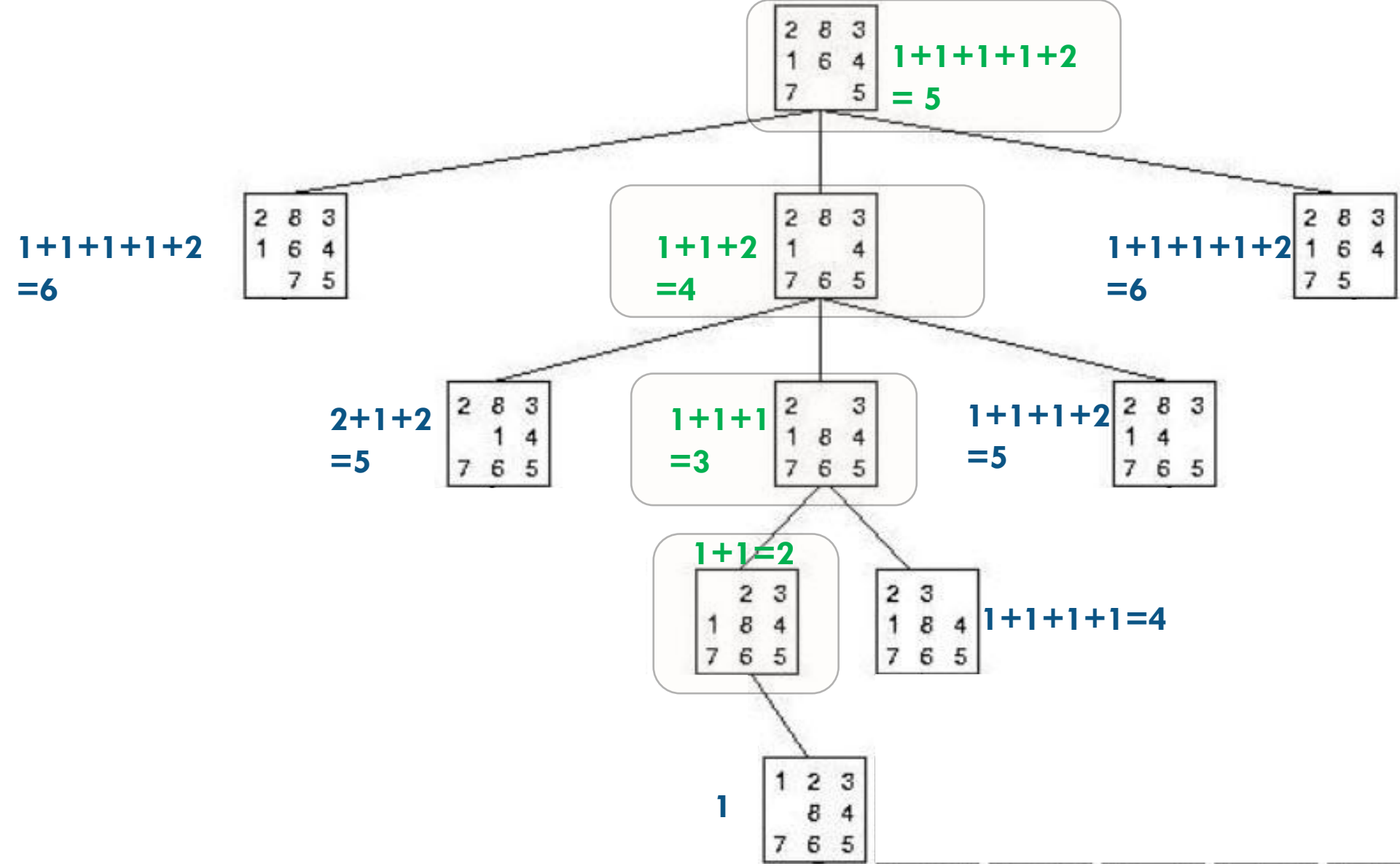
Goal		
1	2	3
8		4
7	6	5



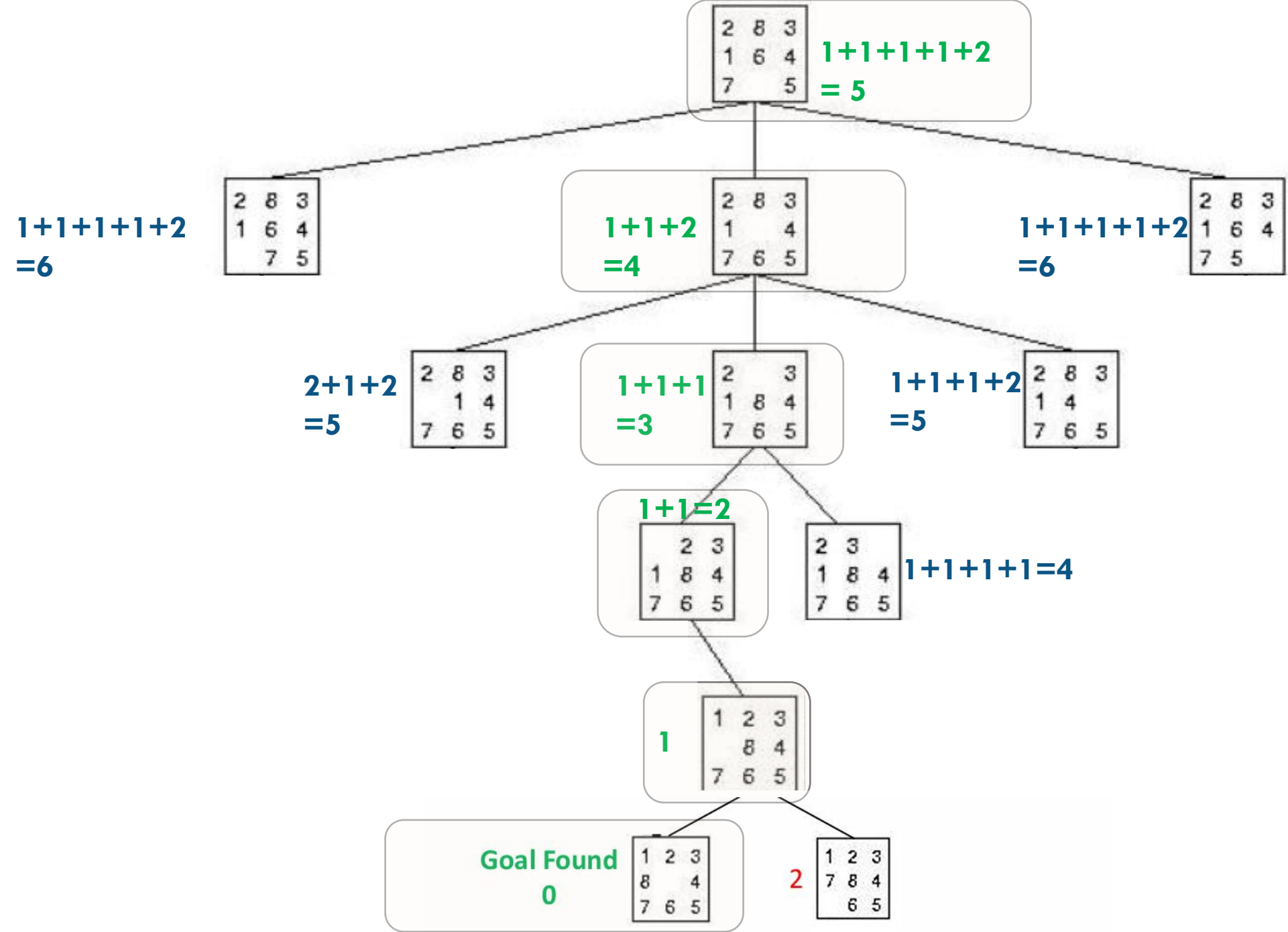
Goal		
1	2	3
8		4
7	6	5



Goal		
1	2	3
8		4
7	6	5

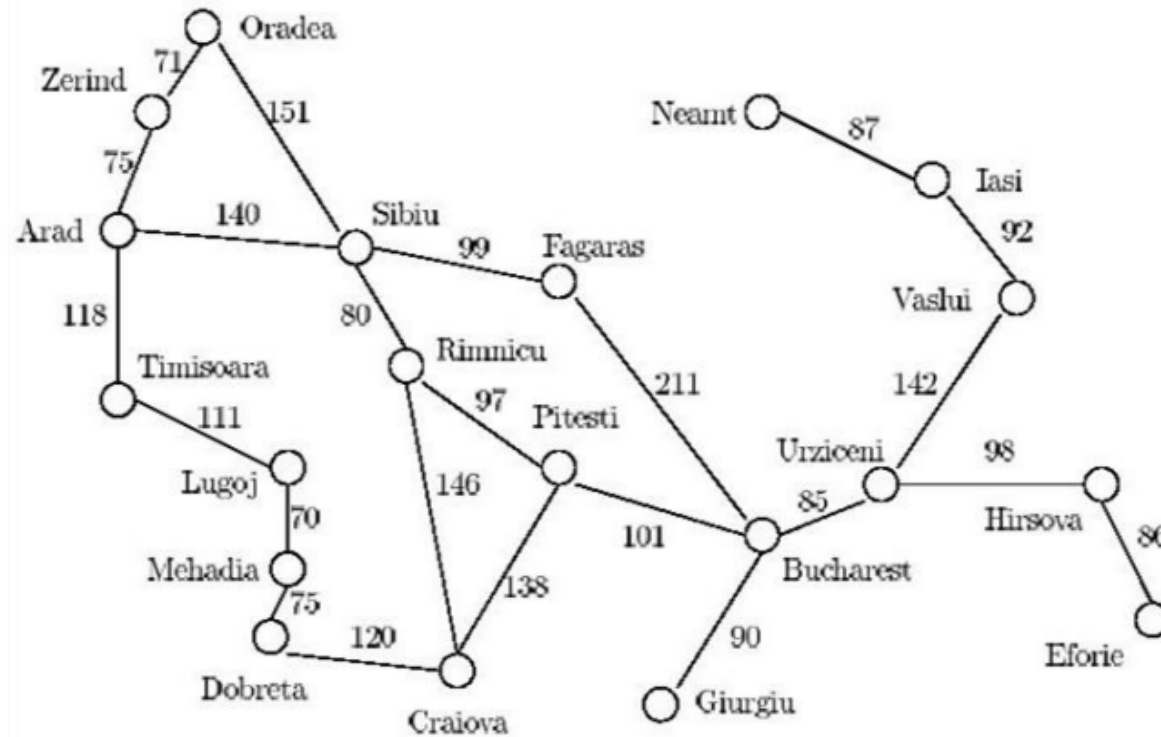


Goal		
1	2	3
8		4
7	6	5



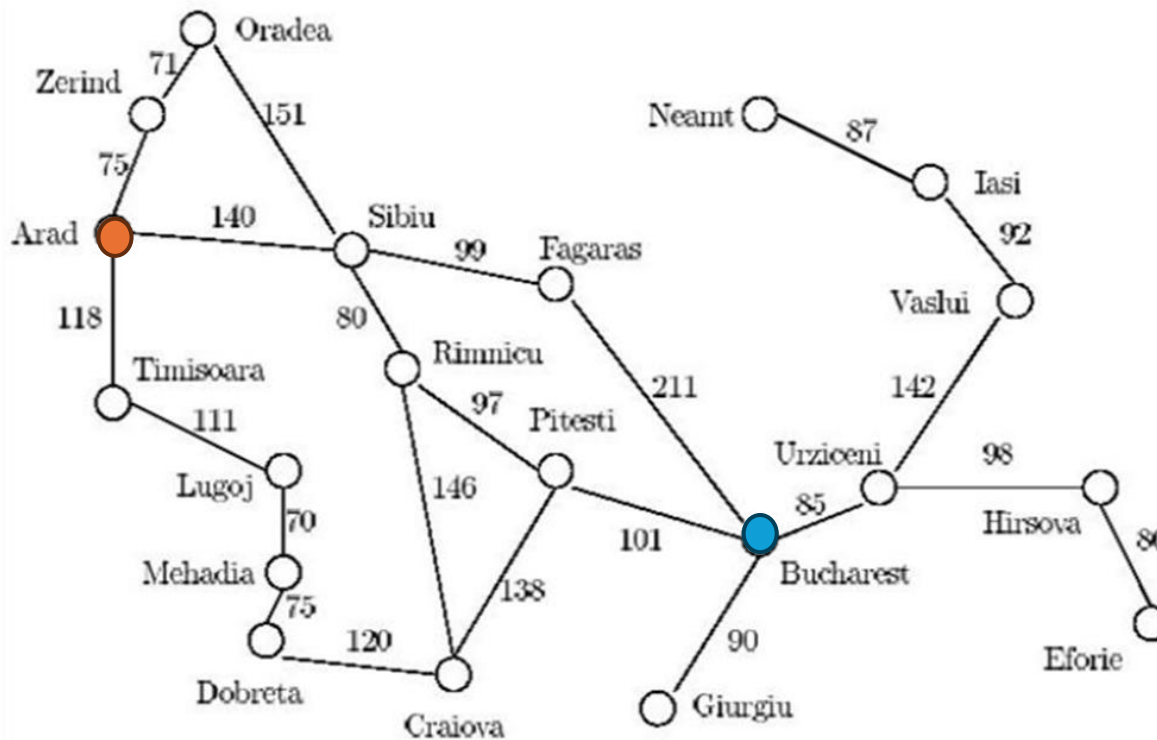
Best First: From Arad/Bucharest

n	H(n)
Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

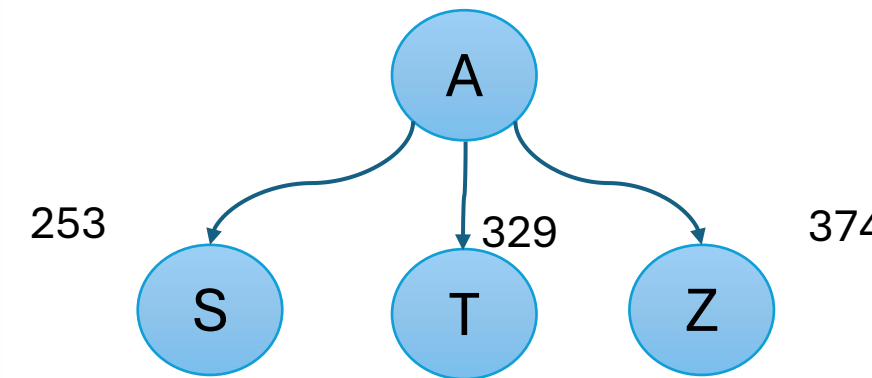


1. Given the **Arad/Bucharest** route finding problem and the heuristics table below, show the list n of visited nodes using the following search techniques:

n	H(n)
Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

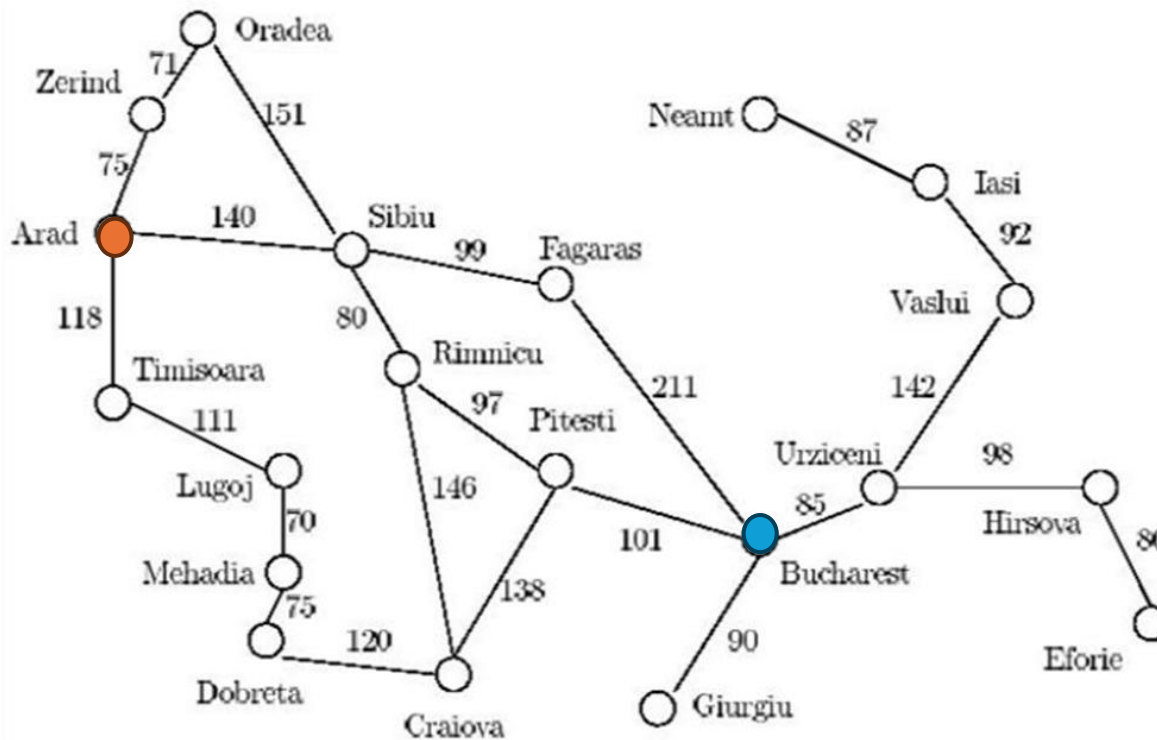


- Best First

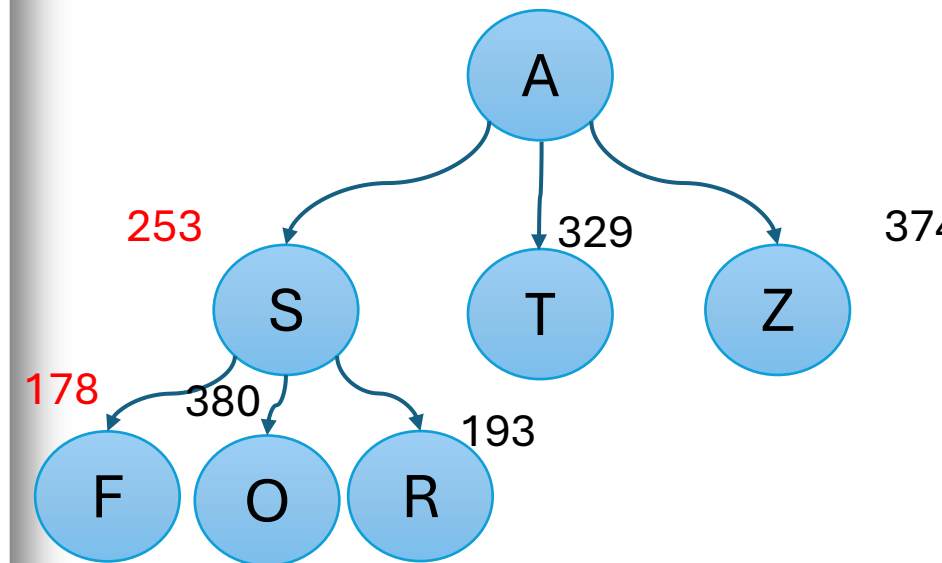


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Sibiu	253
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Urziceni	80
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Zerind	374

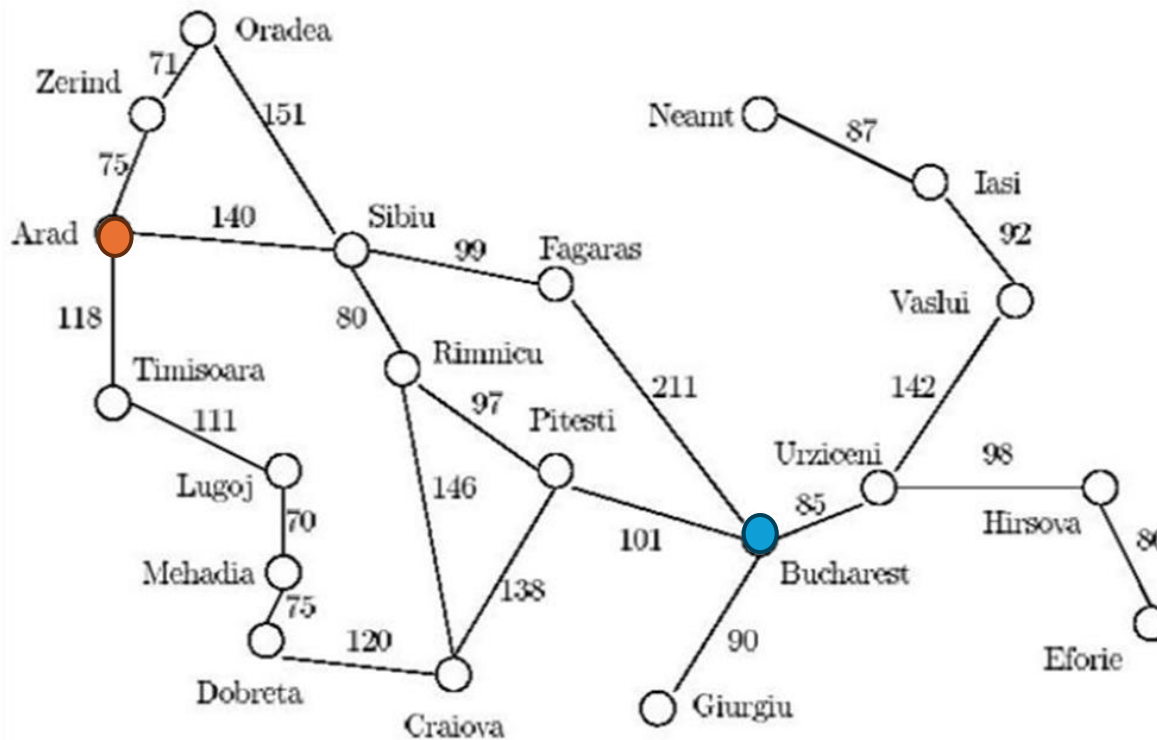


• Best First

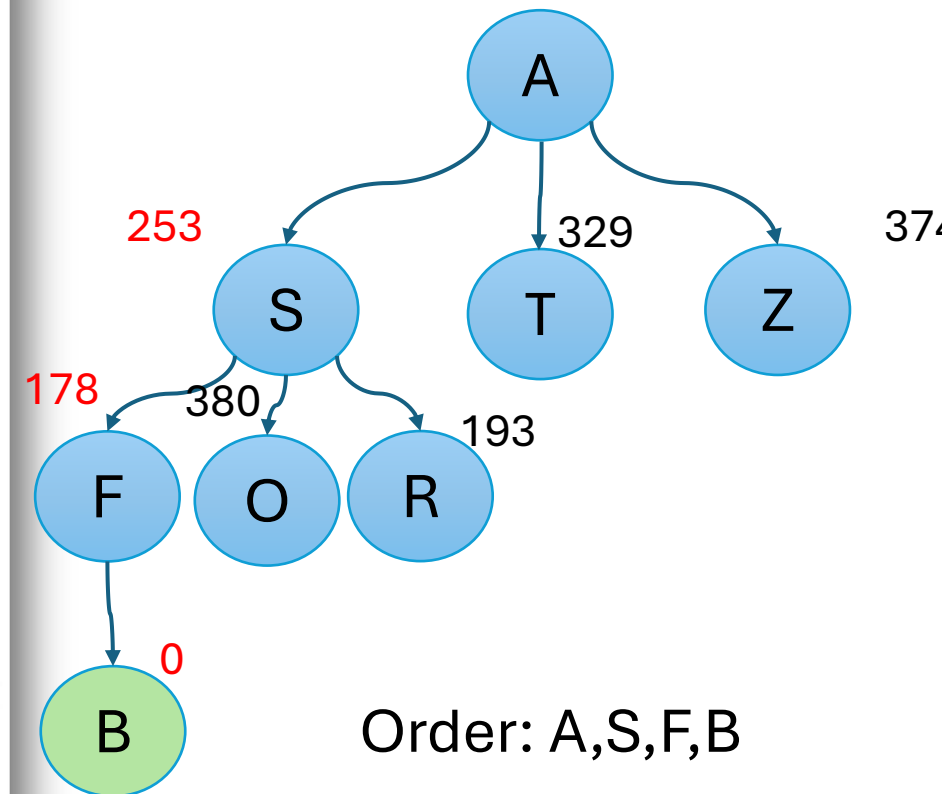


1. Given the **Arad/Bucharest** route finding problem and the heuristics table below, show the list n of visited nodes using the following search techniques:

n	H(n)
Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374



• Best First

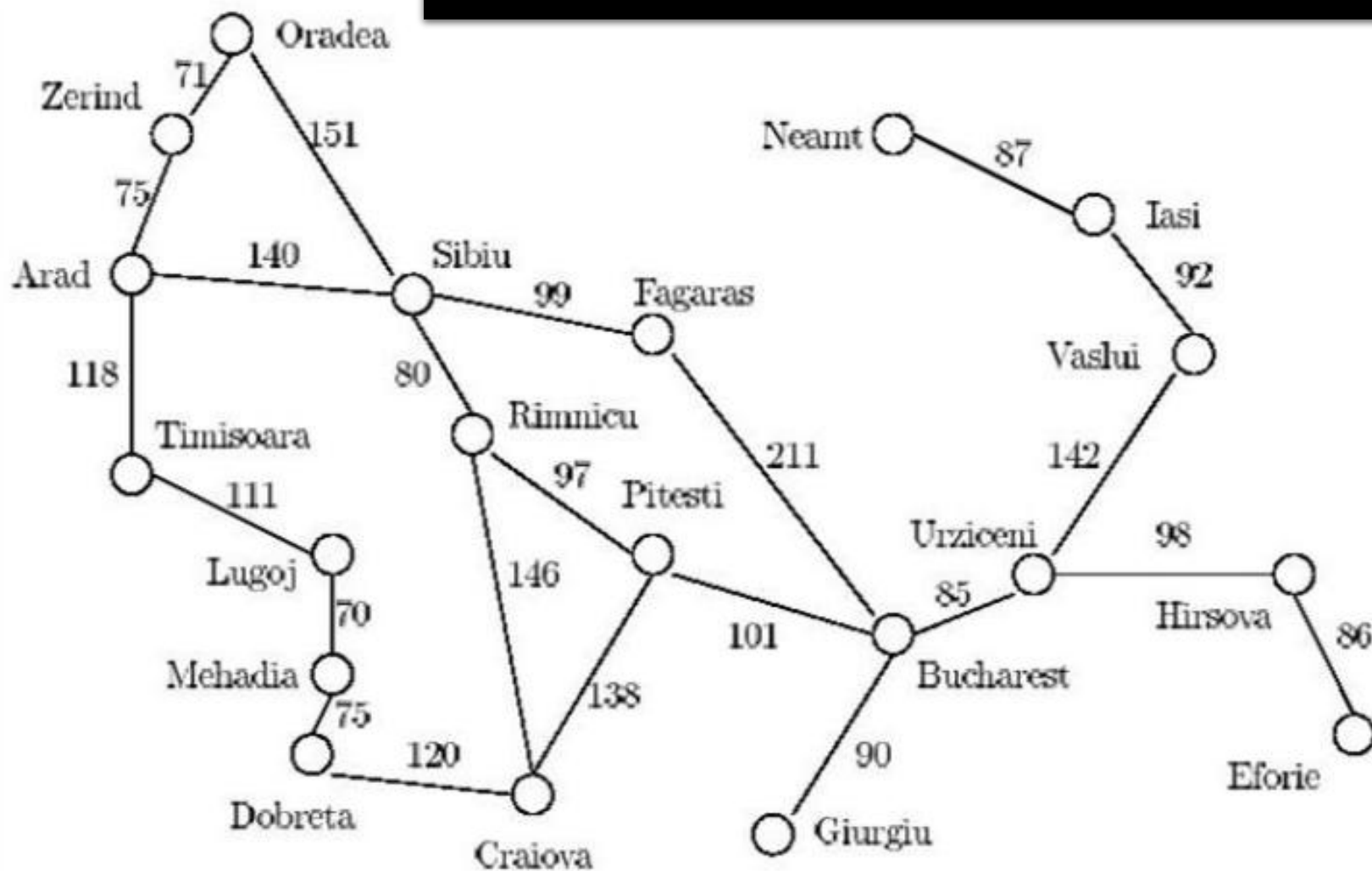


Order: A,S,F,B

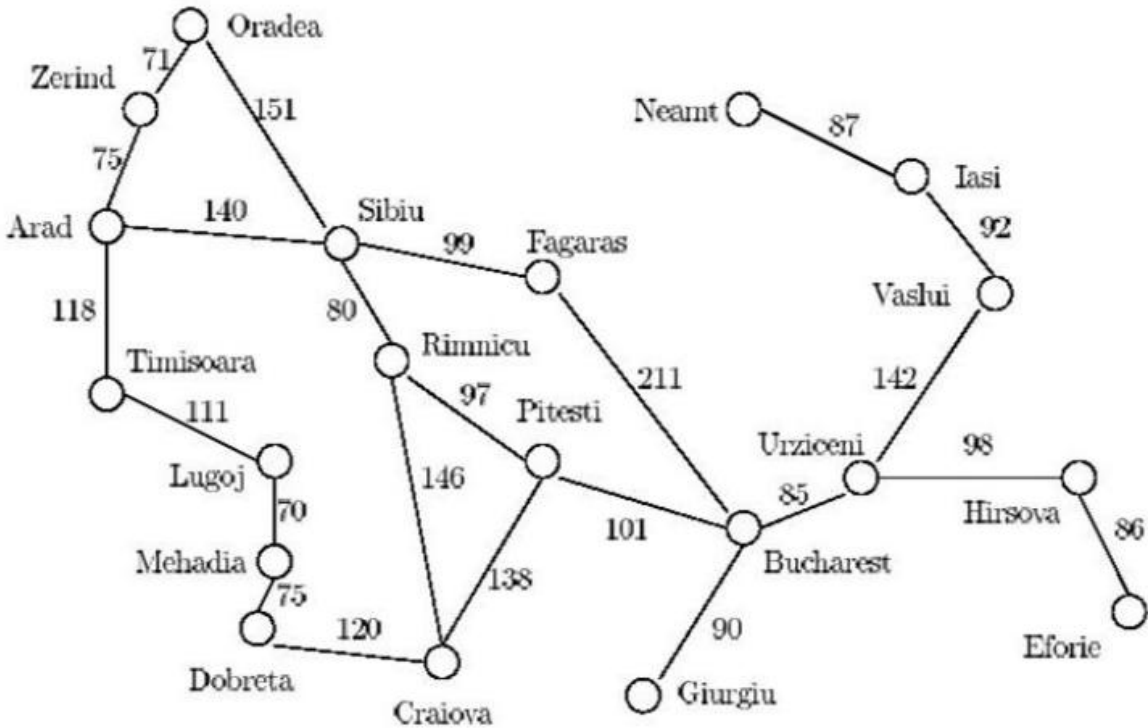
Path: A,S,F,B

n	H(n)
Arad	366
Bucharest	100
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	0
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

Use Best first
Initial: Craiova Goal: Hirsova



n	H(n)
Arad	366
Bucharest	100
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	0
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374



Order: C,P,B,G,U,H
 Path : C,P,B,U,H

